

TechFootPrint

AI-Driven Coding Analytics &
Academic Performance Intelligence

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Dept. of Computer Science & Engineering (AI & ML)

6

Platforms
Integrated

3

User
Roles

AI

Transformer
Recommender

99%

Cluster
Purity

What Colleges Face Today



No Coding Visibility

Faculty see only final submissions. The learning journey — time spent, mistakes, attempts — is completely invisible to them.



Manual Tracking Overhead

Spreadsheets and spot-checks create inconsistent CIE evaluations, massive administrative overhead, and zero scalability.



No Personalized Guidance

Students get generic practice sheets. Without adaptive paths, individual weaknesses go unaddressed until it's too late.



Curriculum–Industry Gap

Slow curriculum updates mean graduates lack the exact DSA and programming skills that companies like Infosys and TCS demand.

TechFootPrint serves

Three Key Stakeholders

Faculty

College Management

Students

FACULTY BENEFITS

Empowering Faculty with Real Intelligence



Real-Time Cohort Dashboard

See every student's solve count, streak, difficulty level and topic coverage — live. No more waiting for end-semester reports.



One-Click CIE Report Generation

Auto-generate Continuous Internal Evaluation reports per student from actual coding activity data — saving hours of manual work.



At-Risk Student Alerts

Automatic flags when a student has low streak, poor topic coverage, or is falling behind peers — so you can intervene early.



Direct Messaging & Discussion

Message students individually or post to discussion forums. Peer collaboration and faculty mentoring in one unified platform.



REAL SCENARIO

Prof. Sontale opens the dashboard before a CIE.



8 students flagged at-risk



Batch avg: 42 problems solved



Weak topic: Dynamic Programming



CIE report downloaded in 1 click



Hours of manual work → 1 click

Strategic Insights for HODs, Deans & Administrators



Curriculum Gap Analysis

Identify which DSA topics are consistently weak across batches. Back curriculum review meetings with evidence, not assumptions.

Data-Driven Decisions



OBE Compliance Tracking

Map coding activity to course outcomes. Generate outcome attainment reports to satisfy NAAC/NBA accreditation requirements automatically.

Accreditation Ready



Placement Readiness Index

Know exactly how ready each student is for campus interviews at Infosys, TCS, and Wipro — before they walk into the placement office.

Industry Aligned



Batch-vs-Batch Benchmarking

Compare performance across batches, sections and semesters. Identify which faculty interventions are working and replicate them.

Institutional Intelligence

Personalized Learning Powered by AI

From Generic Lists to a Personal Learning Path

1. Login
2. AI Analysis
3. Custom Problems
4. Track Progress
5. Placement Ready



AI-Personalised Problem Suggestions

The Transformer engine suggests the next problem based on their history, skill level and weak topics.



Unified Progress Dashboard

All platforms (LeetCode, Codeforces, GFG) in one view — streaks, topics, difficulty distribution.



Placement Readiness Roadmap

Know exactly which topics to focus on to crack Infosys/TCS interviews. AI-generated study plan.



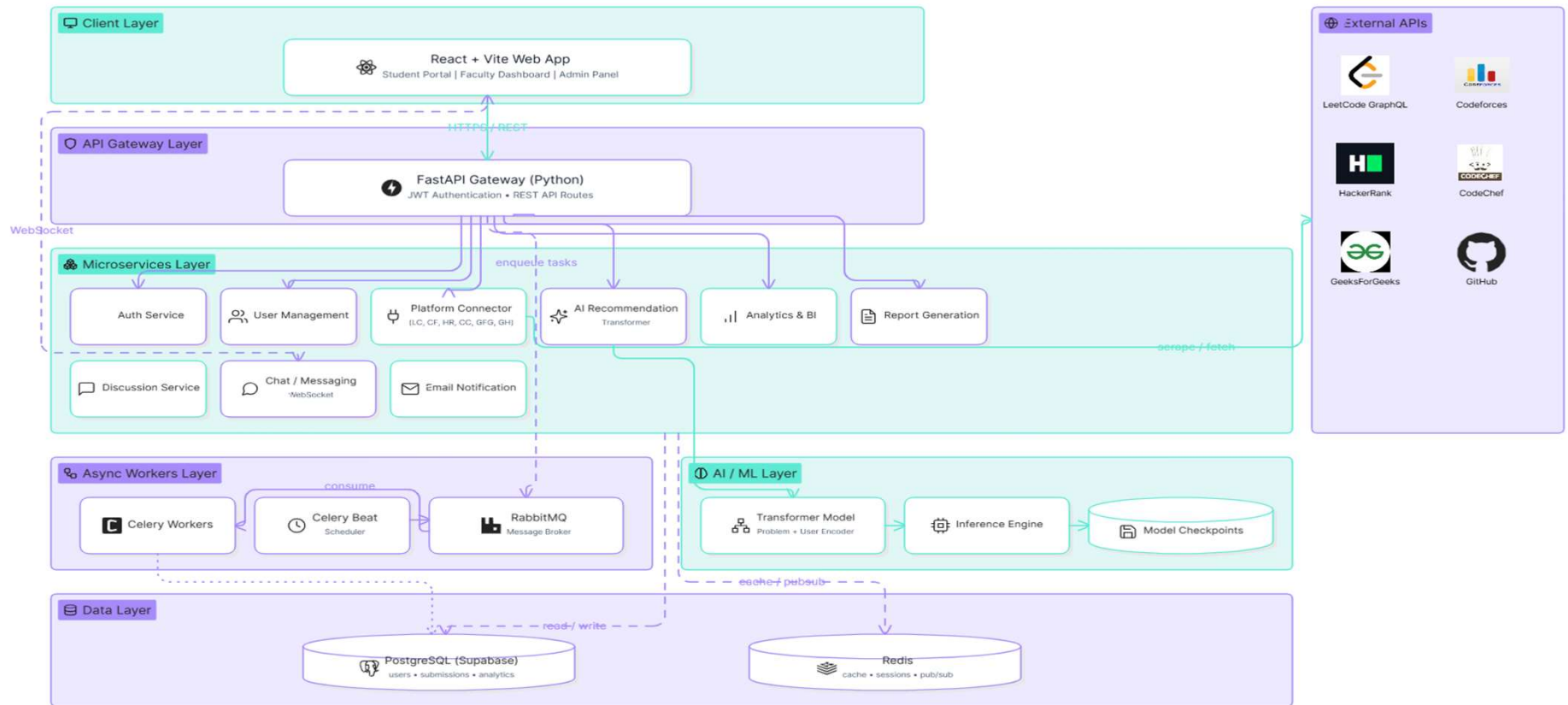
Shareable Public Portfolio

One link that shows total problems solved, platforms used, and achievements — for any recruiter.

SYSTEM ARCHITECTURE

schedule
Real Weights

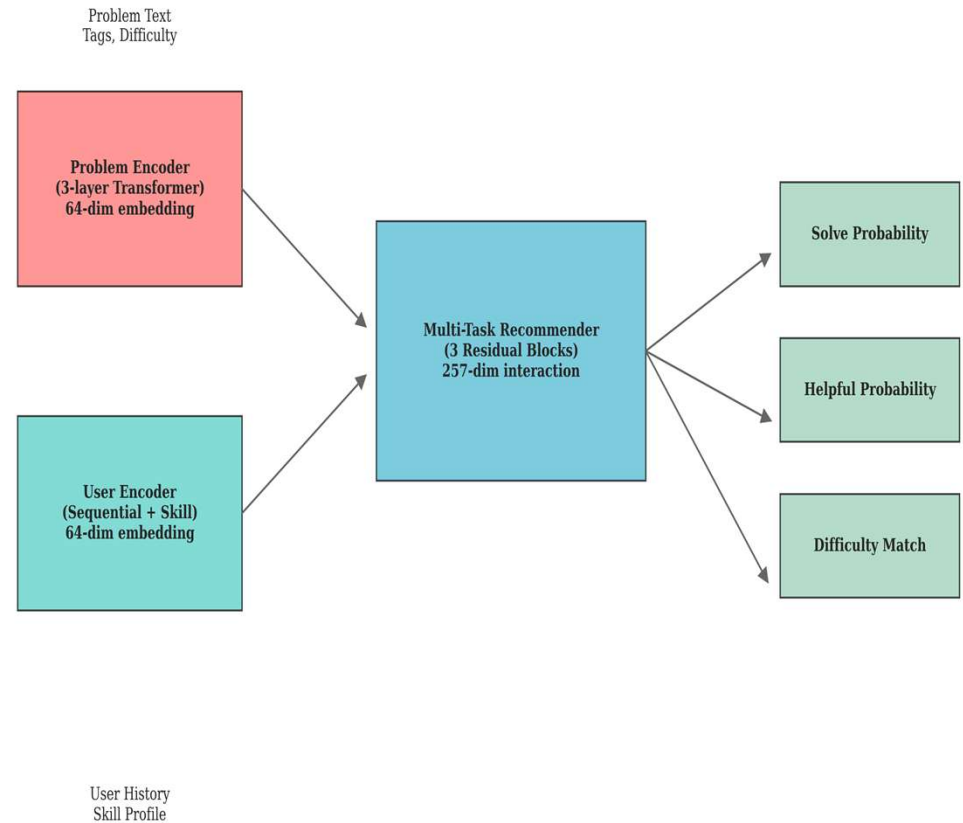
TechFootPrint - System Architecture



AI ENGINE

Self-Trained AI.

- **Trained on 3,433 real LeetCode problems**
Understands problem content, tags, difficulty
- **Learns from each student's history**
Sequential transformer tracks skill progression
- **Predicts what to solve next**
Solve probability · Helpfulness · Difficulty fit
- **11 topic clusters, 99.22% purity**
Semantically organised — problems grouped by concept
- **Ranks problems, not just lists them**
NDCG@10: 75.48% · MRR: 89.66%



PERFORMANCE METRICS

99.22%

Cluster Purity

10 of 11 topic clusters at 100%

89.66%

MRR Score

Relevant problem always near top

75.48%

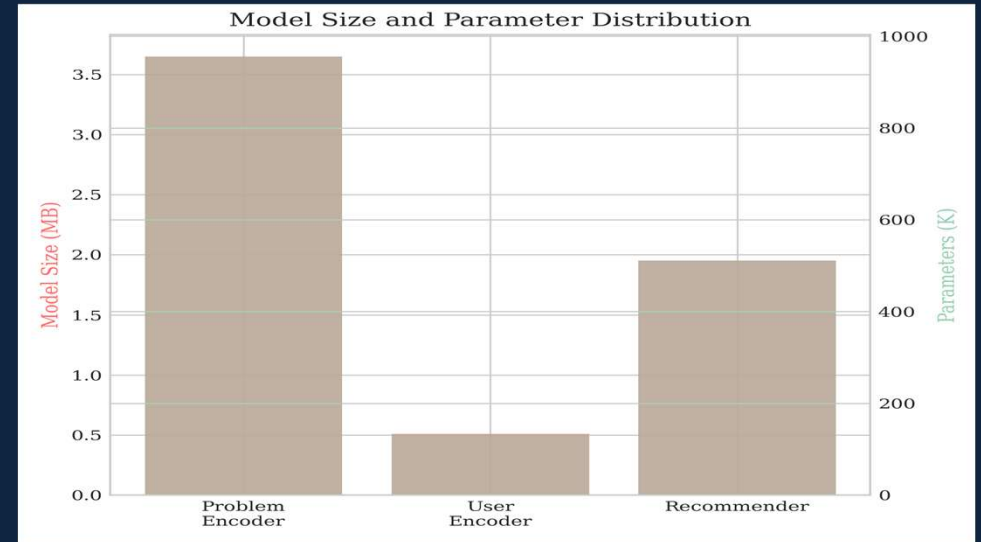
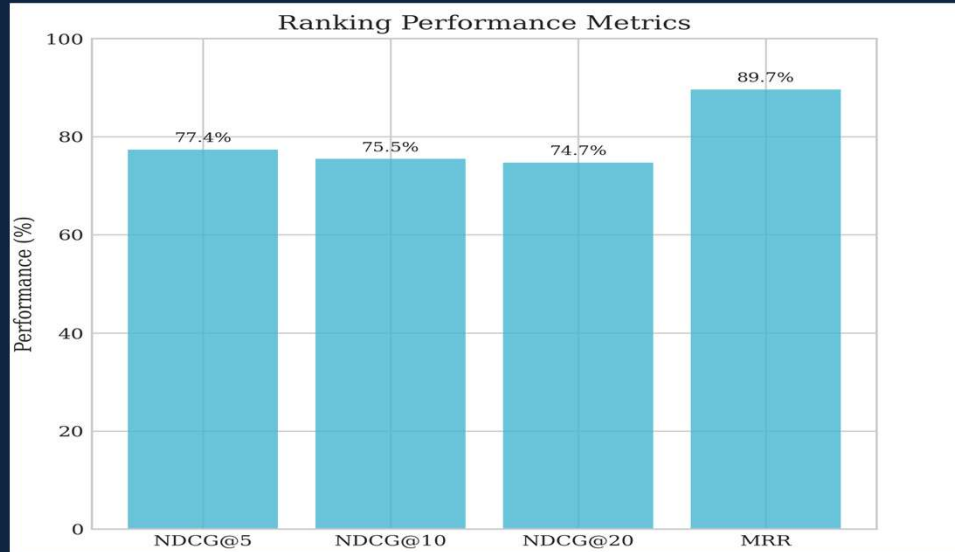
NDCG@10

Strong ranking performance

6.10 MB

Model Size

Compact — deployable on any server



What Is Built & Working Right Now

✓ Implemented & Working

- ✓ Multi-role authentication (Student / Faculty / Admin / Super Admin)
- ✓ Live scraping from LeetCode, Codeforces & GitHub (3 platforms fully live)
- ✓ Background Celery workers — automatic nightly sync with retry & rate-limit handling
- ✓ Analytics dashboard — total solved, streaks, topic distribution, 365-day heatmap
- ✓ Self-trained Transformer DSA Recommender (1.6M params, 6.10 MB, CPU-ready)
- ✓ PostgreSQL schema with Alembic migrations + Redis caching (1hr TTL)
- ✓ Public shareable portfolio with view tracking



In Progress

- ✓ CodeChef, HackerRank, GFG connectors (framework ready, connectors in progress)
- ✓ Faculty BI Dashboard with cohort-level analytics and batch comparisons
- ✓ Automated CIE report PDF generation
- ✓ Real-time direct messaging via WebSockets

Tech Stack at a Glance

- › FastAPI + React 18 + TypeScript
- › PostgreSQL · Redis · RabbitMQ · Celery
- › Docker · Kubernetes · PyTorch (from scratch)

Thank You

Questions Welcome

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3 Roles

Student · Faculty · Admin

6 Platforms

Unified Tracking

99.22%

AI Cluster Purity

Phase 1

Fully Implemented